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BUILT-IN VENTILATION FAN

Abstract

An annular light emitting unit emits light from the inside of a louver toward an outer circumference, with respect to the entire circumference direction. A surface emitting unit emits light in a vertical direction directed opposite to a casing. Light intensity of the surface emitting unit is greater than light intensity of the annular light emitting unit.

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Background/Summary

BACKGROUND

1. Field

[0001] The present disclosure relates to a built-in ventilation fan.

2. Description of the Related Art

[0002] In the main body of a ventilation fan, a ventilation grille is disposed on a surface exposed to the user. The ventilation grille includes a panel having an annular shape, a lamp unit provided in a central part of the panel, and an intake opening provided around the central part (see Patent Literature 1, for example).

[0003] Patent Literature 1: Specification of U.S. Patent Application Publication No. 2018/0180320

[0004] Depending on where the ventilation fan is installed, design may be required for the ventilation fan.

[0005] Meanwhile, also in a situation where design is required, an air flow path needs to be ensured.

SUMMARY

[0006] The present disclosure has been made to solve the problem above, and a purpose thereof is to provide a technology for ensuring an air flow path while improving the design.

[0007] To solve the problem above, a built-in ventilation fan according to one embodiment of the present disclosure includes: a casing that includes a first surface and a second surface different from each other, a suction port provided on the first surface, and a blow-out port provided on the second surface; and a louver that is provided on the upstream side of the suction port and that leads air to the suction port from a direction parallel to the first surface. The louver includes an annular light emitting unit disposed on an inner surface that faces the first surface, and a surface emitting unit disposed on a back surface that faces opposite to the inner surface and that prevents the air from flowing into the suction port from a direction perpendicular to the first surface. The annular light emitting unit emits light from the inside of the louver toward an outer circumference, with respect to the entire circumference direction. The surface emitting unit emits light in the vertical direction directed opposite to the casing. The light intensity of the surface emitting unit is greater than the light intensity of the annular light emitting unit.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] Embodiments will now be described, by way of example only, with reference to the accompanying drawings which are meant to be exemplary, not limiting, and wherein like elements are numbered alike in several Figures, in which:

[0009] FIGS. **1**A-**1**D are diagrams that show a structure of a built-in ventilation fan according to the present embodiment.

[0010] FIGS. **2**A-**2**B are diagrams that show a structure of a casing shown in FIGS. **1**A-**1**D.

[0011] FIGS. **3**A-**3**C are diagrams that show a structure of a louver shown in FIGS. **1**A-**1**D.

[0012] FIG. **4** is an exploded perspective view of the louver shown in FIGS. **3**A-**3**C.

DETAILED DESCRIPTION

[0013] The invention will now be described by reference to the preferred embodiments. This does not intend to limit the scope of the present invention, but to exemplify the invention.

[0014] In the following, the present embodiment will be described with reference to the accompanying drawings. The embodiment described below shows a suitable specific example of the present disclosure. Therefore, the numerical values, shapes, materials, constituting elements, arranged positions and connection forms of the constituting elements, and the like shown in the following embodiment are mere examples and are not intended to limit the present disclosure. Accordingly, among the constituting elements in the following embodiment, a constituting element that is not described in an independent claim, which indicates the most generic concept of the present disclosure, will be described as an arbitrary constituting element. In each drawing, the same reference characters denote substantially the same configurations, and repetitive description will be omitted or simplified.

[0015] FIGS. **1**A-**1**D show a structure of a built-in ventilation fan **1000**. FIG. **1**A is a perspective view of the built-in ventilation fan **1000**. The built-in ventilation fan **1000** includes a casing **100** and a louver **200**. When the casing **100** is embedded in a ceiling surface **10**, the casing **100** is placed in a ceiling space **14**, and the louver **200** is placed in an interior space **12**. At the time, the casing **100** side is referred to as the “upper” side, and the louver **200** side is referred to as the “lower” side. However, the built-in ventilation fan **1000** may also be embedded in a wall surface or the like, instead of the ceiling surface **10**.

[0016] FIG. **1**B is a perspective view of the built-in ventilation fan **1000** viewed from the lower side. FIG. **1**C is a top view of the built-in ventilation fan **1000**. FIG. **1**D is a bottom view of the built-in ventilation fan **1000**. Also, FIGS. **2**A-**2**B show a structure of the casing **100**. FIG. **2**A is a perspective view of the casing **100** viewed from the lower side, and FIG. **2**B is a sectional perspective view taken along line A-A′ in FIG. **2**A. The casing **100** has a hollow box shape constituted by a first surface **110**, a second surface **112**, a third surface **114**, a fourth surface **116**, a fifth surface **118**, and a sixth surface **120**. Since the first surface **110** is a lower surface and the third surface **114** is an upper surface, the first surface **110** and the third surface **114** face in opposite directions. Also, the second surface **112**, fourth surface **116**, fifth surface **118**, and sixth surface **120** are side surfaces, in which the second surface **112** and the fourth surface **116** face in opposite directions, and the fifth surface **118** and the sixth surface **120** face in opposite directions.

[0017] A suction port **130** is provided on the first surface **110**, and a blow-out port **132** is provided on the second surface **112**. The suction port **130** and the blow-out port **132** are openings that each connect the hollow space of the casing **100** and external space. In the hollow space of the casing **100**, a blower **140** is disposed. The blower **140** is, for example, a centrifugal blower that includes a scroll casing and a sirocco fan. The blower **140** is not limited to a centrifugal blower. When the blower **140** operates, air outside the casing **100** is sucked into the hollow space through the suction port **130**, and air in the hollow space is blown out of the casing **100** through the blow-out port **132**. That is, air is sucked into the casing **100** from a direction perpendicular to the first surface **110**, particularly from the lower side to the upper side, and air is blown out of the casing **100** in a direction along the first surface **110**. The blower **140** may not be disposed in the hollow space of the casing **100**. For example, the blower **140** may be disposed at a position posterior to a duct to which the blow-out port **132** is connected. In that case, the casing **100** serves to change the direction in which air flows.

[0018] As shown in FIGS. **1**A-**1**D, the louver **200** is connected to the lower side of the casing **100**, i.e., the upstream side of the suction port **130** of the casing **100**. FIGS. **3**A-**3**C show a structure of the louver **200**. In particular, FIG. **3**A is a perspective view of the louver **200** viewed from the upper side, FIG. **3**B is a perspective view of the louver **200** viewed from the lower side, and FIG. **3**C is a side view of the louver **200**. Further, FIG. **4** is an exploded perspective view of the louver **200**. The louver **200** includes a base **300** disposed on the lower side, and a connection unit **400** disposed on the upper side.

[0019] The base **300** has a quadrangular frustum shape, for example. On the upper side of the base **300**, a base-side inner surface **310** is provided. On the lower side of the base **300**, a back surface **312** facing opposite to the base-side inner surface **310** is provided. The base-side inner surface **310** and the back surface **312** each have a rectangular shape. Although the outer circumference of the back surface **312** will be referred to as an outer circumference edge **302** here, the outer circumference of the base-side inner surface **310** may also be referred to as an outer circumference edge.

[0020] In an inner portion near the center of the base-side inner surface **310**, a projection **322**, which projects annularly in an upward direction from the base-side inner surface **310**, is provided. The upward direction can also be said to be the direction toward the casing **100**. A surface of the projection **322** on the outer circumference side is an outer circumference wall **328**, and the upper edge of the outer circumference wall **328** is a top edge **326**. The top edge **326** of the outer circumference wall **328** has a curved surface shape. Also, outside the projection **322** on the base-side inner surface **310**, multiple, e.g., four, connection protrusions **350** are provided. Each connection protrusion **350** has a columnar shape extending upward from the base-side inner surface **310**.

[0021] The connection unit **400** has an inverted quadrangular frustum shape or a plate shape, for example. On the lower side of the connection unit **400**, a connection unit-side inner surface **410**, which faces the base-side inner surface **310**, is provided. The connection unit-side inner surface **410** has a rectangular shape, and, in the center of the connection unit-side inner surface **410**, a center opening **420** penetrating in a vertical direction is provided. The center opening **420** also has a rectangular shape. Around the center opening **420**, a lattice portion **422** having multiple openings is disposed. The lattice portion **422** reinforces the connection unit **400**, with a lattice shape surrounding the multiple openings. Also, around the center opening **420**, multiple, e.g., four, connection through holes **450** are provided. The size of each connection through hole **450** is adapted to the size of a connection protrusion **350**.

[0022] By inserting the connection protrusions **350** into the connection through holes **450**, the base **300** and the connection unit **400** are connected, with the base-side inner surface **310** and the connection unit-side inner surface **410** separated. At the time, the outer circumference wall **328** of the projection **322** is located inside the lattice portion **422**. Also, the first surface **110** of the casing **100** is connected to the side opposite to the connection unit-side inner surface **410** of the connection unit **400**. Therefore, it can be said that the connection unit **400** connects the base **300** and the casing **100**.

[0023] When the blower **140** in the casing **100** operates in a state where the casing **100**, the base **300**, and the connection unit **400** are connected, air is introduced into a space between the base-side inner surface **310** and the connection unit-side inner surface **410**. The air is directed toward the outer circumference wall **328** of the projection **322** in a direction parallel to the base-side inner surface **310**. The direction parallel to the base-side inner surface **310** can also be said to be a direction parallel to the first surface **110**. The air is directed upward along the outer circumference wall **328** of the projection **322**, passes through the center opening **420** and the lattice portion **422**, and is led to the suction port **130**. To form such a flow of air, the back surface **312** of the base **300** prevents air from flowing into the suction port **130** from a direction perpendicular to the first surface **110**, i.e., prevents flowing in of air from the lower side to the upper side.

[0024] On the back surface **312**, a surface emitting unit **330** of annular shape is disposed. The surface emitting unit **330** emits light, facing downward, i.e., in a vertical direction directed opposite to the casing **100**. Also, on the outer circumference wall **328** of the projection **322**, multiple light emitting units **324** are arranged. The projection **322** and the multiple light emitting units **324** are collectively referred to as an annular light emitting unit **320**. As shown in the arrangement of the projection **322** described above, the annular light emitting unit **320** is disposed, on the base-side inner surface **310**, inside the outer circumference edge **302** of the louver **200**. Since the outer circumference wall **328** of the projection **322** faces toward the outer circumference, the multiple light emitting units **324** emit light toward the outer circumference. Therefore, the annular light emitting unit **320** emits light from the inside of the louver **200** toward the outer circumference, with respect to the entire circumference direction.

[0025] With such arrangement of the annular light emitting unit **320** and the surface emitting unit **330**, the annular light emitting unit **320** serves for indirect illumination, and the surface emitting unit **330** serves for direct illumination. Also, the light intensity of the surface emitting unit **330** is made greater than the light intensity of the annular light emitting unit **320**.

[0026] Inside the annular shape of the projection **322** in the base **300**, a power supply unit **340** is disposed. The power supply unit **340** is connected to the multiple light emitting units **324** and the surface emitting unit **330**. The power supply unit **340** supplies electricity to each of the multiple light emitting units **324** and the surface emitting unit **330** and also controls the light emission of each of the multiple light emitting units **324** and the surface emitting unit **330**. The operation of the power supply unit **340** is not limited thereto.
[0027] In the base **300**, a human detection sensor **342** is disposed inside the annular shape of the projection **322** and closer to the base-side inner surface **310** than the top edge **326** of the projection **322**. The human detection sensor **342** is also exposed in a center portion of the back surface **312**. That is, the human detection sensor **342** is disposed surrounded by the surface emitting unit **330**. The human detection sensor **342** detects the presence of a person on the back surface **312** side, i.e., in the interior space **12**. Since a publicly-known technology may be used for the detection of the presence of a person by the human detection sensor **342**, description thereof is omitted here. The human detection sensor **342** is connected to the power supply unit **340** and outputs the result of detection of a person's presence to the power supply unit **340**. Upon reception of the detection result from the human detection sensor **342**, the power supply unit **340** controls the turning on or off of the multiple light emitting units **324** and also controls the turning on or off of the surface emitting unit **330**.
[0028] According to the present embodiment, the suction port **130** is provided on the first surface **110** of the casing **100**, the louver **200** is provided on the upstream side of the suction port **130**, and air is led to the suction port **130** from a direction parallel to the first surface **110**. Therefore, an air flow path can be ensured. Also, the annular light emitting unit **320** emits light from the inside of the louver **200** toward the outer circumference, with respect to the entire circumference direction, so that indirect illumination can be realized. Since indirect illumination can be realized, the design can be improved. Also, since the light intensity of the surface emitting unit **330** is greater than the light intensity of the annular light emitting unit **320**, brightness can be ensured.
[0029] Also, since the annular light emitting unit **320** is disposed inside the outer circumference edge **302** of the louver **200**, indirect illumination can be realized. Also, since the multiple light emitting units **324** are arranged on the outer circumference wall **328** of the projection **322**, which projects annularly in the direction from the base-side inner surface **310** toward the casing **100**, air can be smoothly directed to the suction port **130**. Also, since the power supply unit **340** is disposed inside the annular shape of the projection **322**, obstruction to a flow of air by the power supply unit **340** can be prevented. Further, since the human detection sensor **342** is disposed inside the annular shape of the projection **322** and closer to the base-side inner surface **310** than the top edge of the projection **322**, obstruction to a flow of air by the power supply unit **340** can be prevented.
[0030] Also, since the center opening **420** and the lattice portion **422**, which allow air to pass through, are arranged in the connection unit **400**, an air flow path can be ensured. Also, since the outer circumference wall **328** of the projection **322** is disposed inside the lattice portion **422**, the base **300** and the connection unit **400** can be combined. Further, since the top edge **326** of the outer circumference wall **328** of the projection **322** has a curved surface shape, air can be smoothly directed to the suction port **130**.
[0031] The outline of one embodiment of the present disclosure is as follows.
Item 1
[0032] A built-in ventilation fan (**1000**), including: [0033] a casing (**100**) that includes a first surface (**110**) and a second surface (**112**) different from each other, a suction port (**130**) provided on the first surface (**110**), and a blow-out port (**132**) provided on the second surface (**112**); and [0034] a louver (**200**) that is provided on the upstream side of the suction port (**130**) and that leads air to the suction port (**130**) from a direction parallel to the first surface (**110**), the louver (**200**) including: [0035] an annular light emitting unit (**320**) disposed on [0036] an inner surface (**310**) that faces the first surface (**110**); and [0037] a surface emitting unit (**330**) disposed on a back surface that faces

opposite to the inner surface (**310**) and that prevents the air from flowing into the suction port (**130**) from a direction perpendicular to the first surface (**110**), [0038] wherein the annular light emitting unit (**320**) emits light from the inside of the louver (**200**) toward an outer circumference, with respect to the entire circumference direction, [0039] wherein the surface emitting unit (**330**) emits light in the vertical direction directed opposite to the casing (**100**), and [0040] wherein light intensity of the surface emitting unit (**330**) is greater than light intensity of the annular light emitting unit (**320**).

Item 2

[0041] The built-in ventilation fan (**1000**) according to Item 1, wherein the annular light emitting unit (**320**) is disposed on the inside of the outer circumference edge (**302**) of the louver (**200**).

Item 3

[0042] The built-in ventilation fan (**1000**) according to Item 1, [0043] wherein the annular light emitting unit (**320**) includes: [0044] a projection (**322**) projecting annularly in a direction from the inner surface (**310**) toward the casing (**100**); and [0045] a plurality of light emitting units (**324**) arranged on an outer circumference wall (**328**) of the projection (**322**), and [0046] wherein the plurality of light emitting units (**324**) emit light toward the outer circumference.

Item 4

[0047] The built-in ventilation fan (**1000**) according to Item 3, [0048] wherein the louver (**200**) includes a power supply unit (**340**) disposed on the inside of the annular shape of the projection (**322**), and [0049] wherein the power supply unit (**340**) supplies electricity to at least the annular light emitting unit (**320**) and controls light emission of the plurality of light emitting units (**324**).

Item 5

[0050] The built-in ventilation fan (**1000**) according to Item 3, [0051] wherein the louver (**200**) includes a human detection sensor (**342**) disposed on the inside of the annular shape of the projection (**322**) and closer to the inner surface (**310**) than a top edge of the projection (**322**), and [0052] wherein the human detection sensor (**342**) detects the presence of a person on the back surface side.

Item 6

[0053] The built-in ventilation fan (**1000**) according to Item 3, [0054] wherein the louver (**200**) includes: [0055] a base (**300**) that includes the annular light emitting unit (**320**) and the surface emitting unit (**330**); and [0056] a connection unit (**400**) that connects the base (**300**) and the casing (**100**), [0057] wherein the connection unit (**400**) includes: [0058] a center opening (**420**) provided in the center; and [0059] a lattice portion (**422**) disposed around the center opening (**420**), [0060] wherein the center opening (**420**) allows the air to pass through, and [0061] wherein the lattice portion (**422**) allows the air to pass through and reinforces the connection unit (**400**).

Item 7

[0062] The built-in ventilation fan (**1000**) according to Item 6, wherein the outer circumference wall (**328**) of the projection (**322**) is disposed on the inside of the lattice portion (**422**).

Item 8

[0063] The built-in ventilation fan (**1000**) according to Item 7, wherein the top edge (**326**) of the outer circumference wall (**328**) of the projection (**322**) has a curved surface shape.

[0064] The present disclosure has been described with reference to an embodiment. The embodiment is intended to be illustrative only, and it will be obvious to those skilled in the art that various modifications to a combination of constituting elements or processes in the embodiment could be developed and that such modifications also fall within the scope of the present disclosure.

Claims

1. A built-in ventilation fan, including: a casing that includes a first surface and a second surface different from each other, a suction port provided on the first surface, and a blow-out port provided on the second surface; and a louver that is provided on the upstream side of the suction port and that leads air to the suction port from a direction parallel to the first surface, the louver including: an annular light emitting unit disposed on an inner surface that faces the first surface; and a surface emitting unit disposed on a back surface that faces opposite to the inner surface and that prevents the air from flowing into the suction port from a direction perpendicular to the first surface, wherein the annular light emitting unit emits light from the inside of the louver toward an outer circumference, with respect to the entire circumference direction, wherein the surface emitting unit emits light in the vertical direction directed opposite to the casing, and wherein light intensity of the surface emitting unit is greater than light intensity of the annular light emitting unit.

2. The built-in ventilation fan according to claim 1, wherein the annular light emitting unit is disposed on the inside of the outer circumference edge of the louver.

3. The built-in ventilation fan according to claim 1, wherein the annular light emitting unit includes: a projection projecting annularly in a direction from the inner surface toward the casing; and a plurality of light emitting units arranged on an outer circumference wall of the projection, and wherein the plurality of light emitting units emit light toward the outer circumference.

4. The built-in ventilation fan according to claim 3, wherein the louver includes a power supply unit disposed on the inside of the annular shape of the projection, and wherein the power supply unit supplies electricity to at least the annular light emitting unit and controls light emission of the plurality of light emitting units.

5. The built-in ventilation fan according to claim 3, wherein the louver includes a human detection sensor disposed on the inside of the annular shape of the projection and closer to the inner surface than a top edge of the projection, and wherein the human detection sensor detects the presence of a person on the back surface side.

6. The built-in ventilation fan according to claim 3, wherein the louver includes: a base that includes the annular light emitting unit and the surface emitting unit; and a connection unit that connects the base and the casing, wherein the connection unit includes: a center opening provided in the center; and a lattice portion disposed around the center opening, wherein the center opening allows the air to pass through, and wherein the lattice portion allows the air to pass through and reinforces the connection unit.

7. The built-in ventilation fan according to claim 6, wherein the outer circumference wall of the projection is disposed on the inside of the lattice portion.

8. The built-in ventilation fan according to claim 7, wherein the top edge of the outer circumference wall of the projection has a curved surface shape.